

A hybrid EEG-fNIRS BCI: motor imagery for EEG and mental arithmetic for fNIRS

M. Jawad Khan¹, Keum-Shik Hong^{1,2,*}, Noman Naseer² and M. Raheel Bhutta²

¹School of Mechanical Engineering, ²Department of Cogno-Mechatronics Engineering,
Pusan National University; 2 Busandaehak-ro, Geumjeong-gu, Busan 609-735, Korea
{jawad, kshong, noman, cogno}@pusan.ac.kr

* Corresponding author

Abstract: In this paper, we have combined electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) to make a hybrid EEG-NIRS based system for brain-computer interface (BCI). The EEG electrodes were placed on the motor cortex region and the NIRS optodes were set on the prefrontal region. The data of four subjects was acquired using mental arithmetic tasks and motor imageries of the left- and right-hand. The EEG data were band-pass filtered to obtain the activity (8~18 Hz). The modified Beer-Lambert law (MBLL) was used to convert the fNIRS data into oxy- and deoxy-hemoglobin (HbO and HbR), respectively. A common threshold between the two modalities was established to define a common resting state. The support vector machines (SVM) was used for data classification. Three control commands were generated using the prefrontal and motor cortex data. The results show that EEG and fNIRS can be combined for better brain signal acquisition and classification for BCI.

Keywords: EEG, fNIRS, hybrid BCI, SVM, Classification

1. INTRODUCTION

A brain-computer interface establishes a link between the brain and computer that can be used to assist patients with locked in syndrome (LIS) [1]. Electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) are the two most common non-invasive techniques used for brain signal acquisition for the purpose of BCI. These modalities have the advantages of being low-cost, portable and having good resistance to noise [2, 3]. EEG acquires brain signals using a neuronal potential from the brain in the form of voltage signals that can be detected by metal electrodes. The detected signals are then amplified and filtered for further analysis. The drawback of using this modality is the inaccuracy in the measured signal due to the deep brain electrical pulses [4-8]. The fNIRS measures the brain signals in the form of concentration changes of oxygenated and deoxygenated hemoglobin (ΔHbO and ΔHbR) using the near-infrared (NIR) light between wavelengths of 600 nm to 1000 nm [9-17]. Both EEG and fNIRS have their own merits and demerits but combining the two enhances the signal strength and classification accuracies. It is done by extracting some relevant feature of EEG and NIRS [18]. The recent researches have shown that the combination of modalities results in better signal acquisition and increased number of control commands that can be used for rehabilitation purposes [19-21].

In this paper, a new approach towards brain signal acquisition and classification is discussed using a hybrid EEG and NIRS system. The signals were acquired from the prefrontal and the motor cortices using mental

arithmetic (MA) and motor imagery, respectively. The acquired signals were filtered and classified using support vector machines (SVM) with an average classification accuracy of 80% for fNIRS and 90% for EEG.

2. METHODS

2.1 Subjects

Four subjects participated in the experiments. The average age of the subjects was 30.5 ± 4.1 years. The subjects had no previous history of any physical, mental or psychological disorder. All subjects had normal or corrected-to-normal vision. The experiments were conducted under the latest declaration of Helsinki and a verbal consent was taken from the subjects before the start of experiments.

2.2 Electrode/ optode placement

The NIRS system (Portalite, Artinis Medical Solutions, Netherlands) was used in the experiment with wavelengths of 760 nm and 830 nm. A single source detector combination was used on the prefrontal cortex. The location of the prefrontal cortex for mental task for each subject was determined before the start of the experiment. Eight EEG Ag/AgCl electrodes (g-MOBilab+ biosignal acquisition device, Christoph Guger, Austria) were placed on the motor cortex region at C3, C4, F3, F4, P3, P4, T3 and T4 with reference at Cz of the International 10-20 system [22]. The optode placement and channel configuration for both EEG and NIRS is shown in Fig. 1.

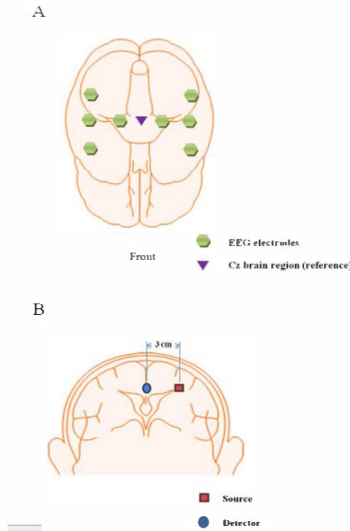


Fig. 1 (A) Electrodes location for EEG, (B) Optodes placement for NIRS.

2.3 Experimental paradigm

A screen was placed approximately at a distance of 70 cm from the subjects. The start of the experiment was a rest period of 20 seconds to establish the baseline for EEG and NIRS. A blue light was displayed on the screen indicating the subjects to imagine the left hand imagery. As the indicated light is turned off the subjects were asked to relax. If the indicated light is green the subject were asked to perform right hand imagery and upon indication of red light, MA was performed. The resting state duration was 20 seconds and activity time was 10 seconds. Two sessions, each consisting of 25 trials was performed on individual subject to obtain the results. The experimental paradigm for a single trial is shown in Fig. 2.

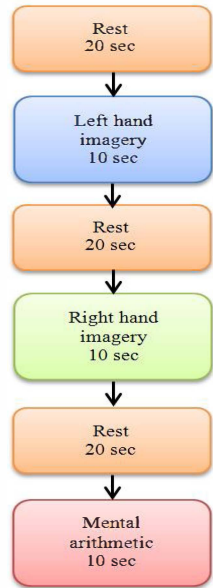


Fig. 2 A single trail consisting of two imagery tasks and one arithmetic task.

2.4 Signal processing

The NIRS data was obtained at 10 Hz sampling rate and the acquired optical density signals were converted into respective changes of ΔHbO and ΔHbR using the modified Beer-Lambert law [14-17]:

$$\begin{bmatrix} \Delta C_{\text{HbO}}(t) \\ \Delta C_{\text{HbR}}(t) \end{bmatrix} = \begin{bmatrix} \alpha_{\text{HbO}}(\lambda_1) & \alpha_{\text{HbR}}(\lambda_1) \\ \alpha_{\text{HbO}}(\lambda_2) & \alpha_{\text{HbR}}(\lambda_2) \end{bmatrix}^{-1} \begin{bmatrix} \Delta A(t, \lambda_1) \\ \Delta A(t, \lambda_2) \end{bmatrix} \frac{1}{l \times d}, \quad (1)$$

where $\Delta A(t; \lambda_j)$ ($j=1,2$) is the unit-less absorbance (optical density) variation of the light emitter of wavelength λ_j , $\alpha_{\text{HbX}}(\lambda_j)$ is the extinction coefficient of HbX in $\mu\text{M}^{-1}\text{mm}^{-1}$, d is the unit-less differential path length factor (DPF), and l is the distance (in millimeters) between emitter and detector. The respiratory and heart beat noises signals are removed from the data using Gaussian low-pass filter and wavelet transform [9-15]. The EEG signals were acquired at a sampling rate of 256 Hz. These were then band-pass filtered between 8-12 Hz and 12-18 Hz to obtain the α - and β -bands for motor imagery tasks. The wavelet transform was used to distinguish the imagery states [2]. The classification signals were carried out using SVM [14].

3. RESULTS

The common threshold established between the modalities is shown in Fig. 3. It has been demonstrated that if both modalities have a common resting state, the system selects a common rest. If only EEG has a resting state then this indicates that NIRS is active and the ΔHbO and ΔHbR are selected as the activated states. On the other hand if only NIRS is at resting state, the EEG signal for left and right imageries are selected for classification. Both states cannot be active at the same time since there is only one indicated color displayed on the screen for a single paradigm. The classification of left and right trials and classification of mental task and rest is shown in Fig. 4. Table 1 shows the classification accuracies of all four subjects for all trials.

Table 1 The classification accuracies of EEG and NIRS signals with a common threshold.

Subject No.	EEG		NIRS
	Right vs. common state	Left vs. Common state	MA vs. Common state
1	92%	92%	80%
2	88%	90%	82%
3	94%	84%	76%
4	90%	94%	82%
Mean	91%	90%	80%

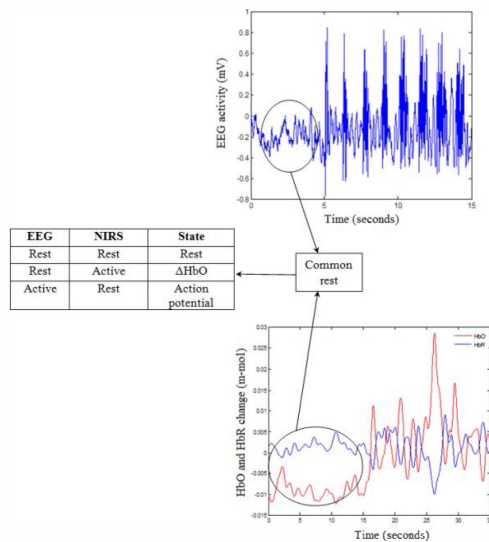


Fig. 3 The common resting state set between EEG and NIRS.

4. DISCUSSION

In the present research and hybrid system based on common resting state was used. Changes corresponding to mental arithmetic task were acquired using NIRS and activity changes due to left and right hand imageries were obtained using EEG. The average classification accuracy for all subjects for NIRS was found to be 80% whereas those for EEG for right and left imageries were 91% and 90%, respectively.

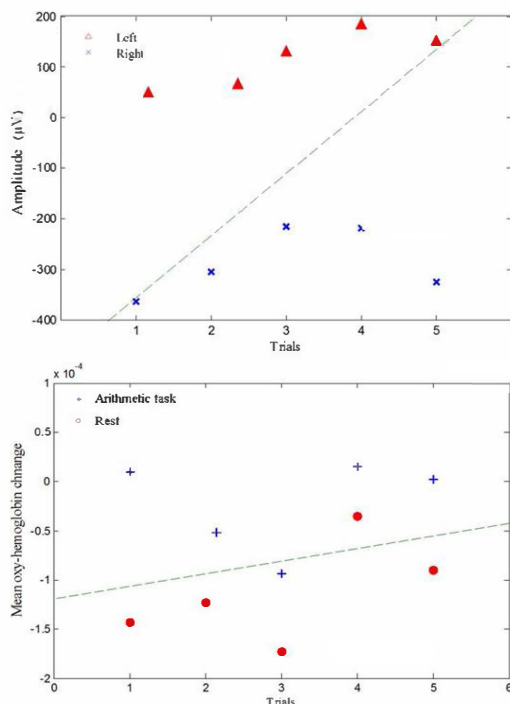


Fig. 4 Left vs. right and mental task vs. rest classification for Subject 2.

The previous studies [14-17] have shown that two control commands can be generated using NIRS using mental arithmetic, mental counting and motor imagery. However using the current configuration the number of control commands can be increased without losing classification accuracy [21].

An online analysis was performed using portable NIRS and EEG systems to obtain three control commands that can be used for BCI and rehabilitation. Therefore this system is suitable for patients with LIS.

In the previous NIRS-BCI studies, light intensity directly is directly used for classification [23], however, converting light intensity into HbO and HbR provides more features for classification. The advantage of using HbO and HbR as features results in increase in the classification accuracy but the command generation time is effected using these features [15].

The limitation of this system is two different time windows for EEG and NIRS. The response of EEG is faster than NIRS, hence the command generation time is faster than NIRS. The command generated by using EEG takes nearly 1 second to execute. In case of NIRS, the control command takes nearly 9 seconds to execute. Here the need is to synchronize the machines to obtain the control signals at the same time interval.

The classification accuracy of the system can further be increased by selecting optimal features for the system. Signal mean was used as feature for classification. As it has been shown that the use of signal mean and signal slope in 2-7 second temporal window within 10 second task period is the best temporal window for classification [15], the same features can be used to further improve the classification of the system.

5. CONCLUSION

In this paper a hybrid system based on EEG and NIRS was used with common resting state to acquire three different control signals from prefrontal and motor cortices. Mental arithmetic task and motor imageries were used to acquire data. Support vector machines (SVM) was used as classifier to generate three control commands. The classification accuracies given in Table 1 indicate that the system is suitable for the purpose of BCI. In the given configurations the system can be used for patients with LIS for the purpose of rehabilitation.

ACKNOWLEDGEMENT

This work was supported by the National Research Foundation of Korea funded by the Ministry of Education, Science and Technology, Korea (grant no. MEST-2012-R1A2A2A01046411).

REFERENCES

- [1] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, "Brain-computer interfaces for communication and control," *Clinical Neurophysiology*, Vol. 113, pp. 767-791, 2002.

- [2] L. F. Nicolas-Alonso and J. Gomez-Gil, "Brain-computer interfaces, a review," *Sensors*, Vol. 12, No. 2, pp. 1211-1279, 2012.
- [3] B. K. Min, M. J. Marzelli, and S.-S. Yoo, "Neuro-imaging based approaches in brain-computer interface," *Trends in Biotechnology*, Vol. 28, No. 11, pp. 552-560, 2010.
- [4] A. Turnip, K.-S. Hong, and M.-Y. Jeong, "Real-time feature extraction of EEG-based P300 using adaptive nonlinear principal component analysis," *Biomedical Engineering Online*, Vol. 10, No. 83, pp. 1-20, 2011.
- [5] T. Ball, M. Kern, I. Mutschler, A. Aertsen, and A. Schulze-Bonhage, "Signal quality of simultaneously recorded invasive and non-invasive EEG," *NeuroImage*, Vol. 46, No. 3, pp. 708-716, 2009.
- [6] T. C. Ferree, M. T. Clay, and D. M. Tucker, "The spatial resolution of scalp EEG," *Neurocomputing*, Vol. 38, pp. 1209-1216, 2001.
- [7] G. Lucci, M. Berchicci, D. Spinelli, F. Taddei, and, F. Di Russo, "The Effects of Aging on Conflict Detection," *PLoS One*, Vol. 8, No. 2, e00565669, 2013.
- [8] F. Barlaam, M. Descoins, O. Bertrand, T. Hasbroucq, F. Vidal, C. Assaïante, and C. Schmitz, "Time-frequency and ERP analysis of EEG to characterize anticipatory postural adjustments in bimanual load lifting task," *Frontiers in Human Neuroscience*, Vol. 5, e00163, 2011.
- [9] M. A. Kamran, and K.-S. Hong, "Linear parameter varying model and adaptive filtering technique for detecting neuronal activities: an fNIRS study," *Journal of Neural Engineering*, Vol. 10, No. 5, e056002, 2013.
- [10] M. R. Bhutta, K.-S. Hong, B.-M. Kim, M. J. Hong, Y.-H. Kim, and S.-H. Lee, "Note: Three wavelength near-infrared spectroscopy system for compensating the light absorbance by water," *Review of Scientific Instruments*, Vol. 85, No. 2, e026111, 2014.
- [11] H. Santosa, M. J. Hong, S.-P. Kim, and K.-S. Hong, "Noise reduction in functional near-infrared spectroscopy signals by independent component analysis," *Review of Scientific Instruments*, Vol. 84, No. 7, e073106, 2013.
- [12] X.-S. Hu, K.-S. Hong, and S. S. Ge, "fNIRS-based online deception decoding," *Journal of Neural Engineering*, Vol. 9, No. 2, e026012, 2012.
- [13] K.-S. Hong, and H.-D. Nguyen, "State-space models of impulse hemodynamic responses over motor somatosensory, and visual cortice," *Biomedical Optics Express*, Vol. 5, No. 6, pp. 1778-1798, 2014.
- [14] N. Naseer, M. J. Hong, and K.-S. Hong, "Online binary decision decoding using functional near-infrared spectroscopy for the development of brain-computer interface," *Experimental Brain Research*, Vol. 232, No. 2, pp. 555-564, 2014.
- [15] N. Naseer, and K.-S. Hong, "Classification of functional near-infrared spectroscopy signals corresponding to right- and left-wrist motor imagery for development of a brain-computer interface," *Neuroscience Letters*, Vol. 553, pp. 84-89, 2013.
- [16] S. M. Coyle, T. E. Ward, and C. M. Markham, "Brain-computer interface using a simplified functional near-infrared spectroscopy system," *Journal of Neural Engineering*, Vol. 4, No. 3, pp. 219-226, 2007.
- [17] R. Sitaram, H. Zhang, C. Guan, M. Thulasidas, Y. Hoshi, A. Ishikawa, K. Shimizu, and N. Birbaumer, "Temporal classification of multichannel near-infrared spectroscopy signals of motor imagery for developing a brain-computer interface," *NeuroImage*, Vol. 34, No. 4, pp. 1416-1427, 2007.
- [18] S. Fazli, J. Mehnert, J. Steinbrink, G. Curio, A. Villringer, K.-R. Müller, and B. Blankertz, "Enhanced performance by a hybrid NIRS-EEG brain computer interface," *NeuroImage*, Vol. 59, No. 1, pp. 512-529, 2012.
- [19] H.-J. Hwang, D. H. Kim, C.-H. Han, and C.-H. Im, "A new dual frequency stimulation method to increase the number of visual stimuli for multi-class SSVEP-based brain-computer interface (BCI)," *Brain Research*, Vol. 1515, pp. 66-77, 2013.
- [20] Y. Li, J. Pan, F. Wand, and Z. Yu, "A hybrid BCI system combining P300 and SSVEP and its applications to wheelchair control," *IEEE Transactions on Biomedical Engineering*, Vol. 60, No. 11, pp. 3156-3166, 2013.
- [21] M. J. Khan, M. J. Hong, and K.-S. Hong, "Decoding of four movement directions using hybrid NIRS-EEG brain-computer interface," *Frontiers in Human Neuroscience*, Vol. 8, e00244, pp. 1-10, 2014.
- [22] V. Jurcak, D. Tsuzuki, and I. Dan, "10/20, 10/10, and 10/5 system revisited: Their validity as head-surface-based positioning system," *Neuroimage*, Vol. 34, No. 4, pp. 1600-1611, 2007.
- [23] M. Naito, Y. Michioka, K. Ozawa, Y. Ito, M. Kiguchi and T. Kanazaw, "A communication means for totally locked-in ALS patients based on changes in cerebral blood volume measured with near-infrared light," *IEICE Transaction on Information and Systems*, Vol. 90, pp. 1028-1037, 2007.